Hyperuniformity in Chemotactic Active Matter

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1 Models

1.1 General Model

$$\dot{\mathbf{r}}_i\left(t\right) = v\mathbf{p}\left(\theta_i\right) + \sum_{j\in A_i^{1,2}} \mathbf{I}\left(\Delta\mathbf{r}_{ij}\right) , \tag{1a}$$

$$\dot{\theta}_i\left(t\right) = \omega_i + G\left(\mathbf{r},\theta,c\right) + \sum_{j\neq i} H\left(\Delta\theta_{ij},\Delta\mathbf{r}_{ij}\right) , \tag{1b}$$

$$\dot{c}\left(\mathbf{r},t\right) = D\nabla^2 c + F\left(c\right)\sum_{j=1}\delta\left(\mathbf{r}-\mathbf{r}_i\right) , \tag{1c}$$

for $i = 1, 2, \cdots, N$. Here, $\mathbf{r}_i$ is the position of the i-th particle, θ_i is the orientation of the i-th particle, v is the self-propulsion velocity, $\mathbf{p}\left(\theta_i\right) = \left(\cos\theta_i, \sin\theta_i\right)$ is the unit vector pointing in the direction of the i-th particle, ω_i is the natural frequency of the i-th particle, $G\left(\mathbf{r},\theta,c\right)$ is the coupling function between particles and chemical fields, $H\left(\Delta\theta_{ij},\Delta\mathbf{r}_{ij}\right)$ is the coupling function between partials, $c\left(\mathbf{r},t\right)$ is the chemical concentration, D is the diffusion coefficient, $F\left(c\right)$ is the production rate of the chemical field, $A_i^{1,2} = \{j \mid r_c \geqslant |\mathbf{r}_j - \mathbf{r}_i|\}$, $\Delta\mathbf{r}_{ij} = \mathbf{r}_j - \mathbf{r}_i$, $\Delta\theta_{ij} = \theta_j - \theta_i$, $\mathbf{I}\left(\Delta\mathbf{r}_{ij}\right) = \frac{\Delta\mathbf{r}_{ij}}{|\Delta\mathbf{r}_{ij}|^2}$. The natural frequencies ω_i are distributed with following two cases:

1.2 Chemotactic Lotka-Volterra Model

Let $F_1\left(c_1,c_2\right) = k_1c_1\left(1-c_2\right)$ and $F_2\left(c_1,c_2\right) = k_2c_2\left(c_1-1\right)$, where k_1, k_2 are constants.

$$\dot{\mathbf{r}}_i^{1,2} = v\mathbf{p}\left(\theta_i^{1,2}\right) - \sum_{j\in A_i^{1,2}} \mathbf{I}_{ij}^{1,2} , \tag{2a}$$

$$\dot{\theta}_i^{1,2} = \omega + |\nabla c_{1,2}|\sin\left(\varphi_{c_{1,2}} - \theta_i^{1,2}\right) + F\left(\theta,\mathbf{r}\right) , \tag{2b}$$

$$\dot{c}_1 = D_1\nabla^2 c_1 + k_1c_1\left(1-c_2\right)\sum_{j=1}^{N}\delta\left(\mathbf{r}-\mathbf{r}_j^1\right) , \tag{2c}$$

$$\dot{c}_2 = D_2\nabla^2 c_2 + k_2c_2\left(c_1-1\right)\sum_{j=1}^{N}\delta\left(\mathbf{r}-\mathbf{r}_j^2\right) , \tag{2d}$$

where $\mathbf{r}_i^{1,2}$ and $\theta_i^{1,2}$ are the position and orientation of the i-th particle of species 1 and 2, respectively. v is the self-propulsion speed, ω is the intrinsic angular velocity, and $\mathbf{p}(\theta) = (\cos\theta, \sin\theta)$ is the unit vector in the direction of θ. The term $\sum_{j \in A_i^{1,2}} \mathbf{I}_{ij}^{1,2}$ represents the interaction forces between particles, where $A_i^{1,2}$ denotes the set of neighboring particles that interact with particle i of species 1 or 2. The function $F(\theta, \mathbf{r})$ represents any additional torque that may depend on the orientation and position of the particles. The equations for c_1 and c_2 describe the diffusion and production of the chemical fields by the particles. The parameters D_1 and D_2 are the diffusion coefficients for the chemical fields, and the terms involving k_1 and k_2 represent the production of the chemicals by the particles, which depends on the local concentrations of the chemicals.